EE4-45
EE9-SO22
EE9-FPN2-09

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Wednesday, 29 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A. Manikas

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

The linear B-spline is given by

$$\varphi(t) = \begin{cases} 1-|t|, & |t|<1 \\ \\ 0, & \text{otherwise.} \end{cases}$$

The Questions

1. Consider the system shown in Figure 1a



Figure 1a: An interpolator.

(a) What does the product filter $P(z) = H(z)G(z)$ have to satisfy in order to have perfect reconstruction such that $y[n] = x[n]$? [5]

(b) Assume that $H(z) = (z^{-2} + z^{-1} + 1 + z + z^2)$. Find the shortest symmetric filter $G(z)$ such that perfect reconstruction is achieved. [5]

(c) Now assume that $H(z) = (1+z+z^2+z^3)$, find the shortest $G(z)$ such that $Y(z) = 0$. [5]

(d) Consider the multi-rate system of Fig. 1b. Assume that $H_1(z)$ and $H_2(z)$ have only even-numbered coefficients: $h_1[n] = (..., h_1[0], 0, h_1[2], 0, ...)$ and likewise for $h_2[n]$.

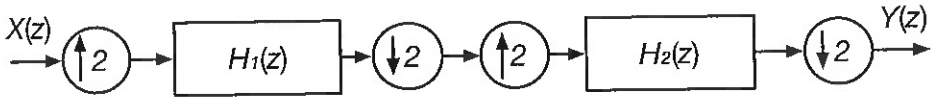


Figure 1b: Multi-rate System

i. Express $Y(z)$ as a function of $X(z)$ and the filters. [5]

ii. Assume that $H_1(z)$ and $H_2(z)$ are both ideal low-pass filters with cut-off frequency $\pi/2$, can you claim that $Y(z) = X(z)$? Justify your answer. [5]

2. Consider a biorthogonal scaling function $\varphi(t)$ and its dual $\tilde{\varphi}(t)$. The two functions satisfy the following two-scale equations:

$$\varphi(t) = \sqrt{(2)} \sum_n g_0[n]\varphi(2t-n)$$

and

$$\tilde{\varphi}(t) = \sqrt{(2)} \sum_n h_0[n]\tilde{\varphi}(2t-n).$$

(a) Show that the biorthogonality condition $\langle \tilde{\varphi}(t), \varphi(t-n)\rangle = \delta[n]$ implies that $\langle h_0[k+2n], g_0[k]\rangle = \delta[n]$. [5]

(b) Assume that $\varphi(t)$ is a linear B-spline given by

$$\varphi(t) = \begin{cases} 1-|t|, & |t|<1 \\ 0, & \text{otherwise.} \end{cases}$$

In this case, the z-transform of $g_0[n]$ is

$$G_0(z) = \frac{1}{2\sqrt{2}}(z+2+z^{-1}).$$

Using the following half-band filter

$$P(z) = H_0(z)G_0(z) = \frac{1}{16}(1+z)^2(1+z^{-1})^2(-z+4-z^{-1}).$$

sketch and dimension the corresponding wavelet

$$\psi(t) = \sqrt{(2)} \sum_n (-1)^{n-1} h_0[1-n]\varphi(2t-n).$$

[5]

(c) The dual of $\psi(t)$ is given by

$$\tilde{\psi}(t) = \sqrt{(2)} \sum_n (-1)^{n-1} g_0[1-n]\tilde{\varphi}(2t-n).$$

How many vanishing moments has $\tilde{\psi}(t)$? Justify your answer. [5]

Question continues on next page.

(d) A function $f(t)$ uniformly Lipshitz in $[a,b]$ with Lipshitz coefficients $\alpha = 1.6$ is decomposed using $\psi(t)$:

$$f(t) = \sum_n \sum_m \langle f(t), \tilde{\psi}_{m,n}(t) \rangle \psi_{m,n}(t).$$

Show that the wavelet coefficients in the cone of influence of $t_0 \in [a,b]$ decay as follows: $\langle f(t), \tilde{\psi}_{m,n}(t) \rangle \sim C_1 2^{m(\alpha+1/2)}$ where C_1 is a constant. [5]

(e) Assume now that $\varphi(t)$ is a zero-order B-spline given by

$$\varphi(t) = \begin{cases} 1, & t \in [0,1) \\ 0, & \text{otherwise.} \end{cases}$$

In this new case, the z-transform of $g_0[n]$ is

$$G_0(z) = \frac{1}{\sqrt{2}}(1 + z^{-1}).$$

Using the following half-band filter

$$P(z) = \frac{1}{2}(1+z)(1+z^{-1}),$$

i. Sketch and dimension the corresponding wavelet

$$\psi(t) = \sqrt{(2)} \sum_n (-1)^{n-1} h_0[1-n] \varphi(2t-n).$$

[3]

ii. State the number of vanishing moments of this new wavelet. [2]

3. You are asked to approximate a sequence $x[n]$ with a sequence $y_0[n]$ having the following form:

$$y_0[n] = \sum_{k\in\mathbb{Z}} \alpha_0[k]\phi_0[n-3k]$$

where $\phi_0[n]$ is a length-3 sequence with z-transform given by

$$\Phi_0(z) = (1 + z^{-1} + z^{-2})/\sqrt{3}.$$

(a) Draw a block-diagram of a system having $\alpha_0[n]$ as input and $y_0[n]$ as output. [6]

(b) We compute $\alpha_0[n] = (x[3n] + x[3n+1] + x[3n+2])/\sqrt{3}$. Draw a block diagram of a system having $x[n]$ as input and $y_0[n]$ as output. Explain what that system does. [6]

(c) You wish to find two sequences $y_1[n]$ and $y_2[n]$ which, when added to $y_0[n]$, exactly recovers $x[n]$. That is, $y_0[n] + y_1[n] + y_2[n] = x[n]$. The sequences $y_i[n]$ $i = 1, 2$ can be written as follows:

$$y_i[n] = \sum_{k\in\mathbb{Z}} \alpha_i[k]\phi_i[n-3k], \text{ for } i = 1, 2.$$

i. Find the length-3 sequences $\phi_1[n]$ and $\phi_2[n]$. You want the two sequences to be causal. You also want $\phi_1[0] = \phi_1[2]$ and $\phi_2[0] = -\phi_2[2]$. [8]

ii. Give an expression for the coefficients $\alpha_1[k]$ and $\alpha_2[k]$ in terms of $x[n]$. [5]

4. Consider the two-channel filter bank of Figure 3a.

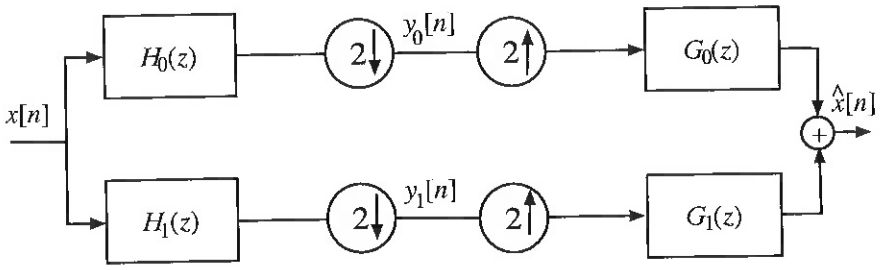


Figure 3a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [6]

(b) Assume that $P(z) = H_0(z)G_0(z) = \frac{1}{2}(1+z^{-1})(1+z)$ and assume that $G_0(z) = (z+2+z^{-1})/(2\sqrt{2})$. Find the three filters $H_0(z)$,$H_1(z)$ and $G_1(z)$ that would lead to a perfect reconstruction filter-bank. [6]

(c) Consider the equivalent filter

$$G_0^{(i)}(z) = \prod_{k=0}^{i-1} G_0(z^{2^k})$$

obtained by iterating filter $G_0(z)$ in the filter bank decomposition i times. Consider the function

$$\varphi^{(i)}(t) = 2^{i/2} g_0^{(i)}[n], \qquad n/2^i \le t < (n+1)/2^i.$$

We know that $\varphi(t) = \lim_{i\to\infty} \varphi^{(i)}(t)$ is the linear spline. Sketch and dimension the wavelet

$$\psi(t) = \sqrt{2}\sum_n (-1)^{n-1} h_0[1-n]\varphi(2t-n).$$

[3]

Question continues on next page.

(d) Now consider the equivalent filter

$$H_0^{(i)}(z) = \prod_{k=0}^{i-1} H_0(z^{2^k})$$

obtained by iterating filter $H_0(z)$ in the filter bank decomposition i times. Consider the function

$$\tilde{\varphi}^{(i)}(t) = 2^{i/2} h_0^{(i)}[n], \qquad n/2^i \leq t < (n+1)/2^i.$$

Find the expression for $\tilde{\varphi}(t) = \lim_{i\to\infty} \tilde{\varphi}^{(i)}(t)$. [3]

(e) Consider the two-channel filter bank of Figure 3a without downsamplers and upsamplers. Such a filter bank is called a *Nonsubsampled* filter bank. Assume that $H_0(z) = 1$ and $G_0(z) = z^{-1} + 4 + z$, calculate $\{H_1(z), G_1(z)\}$ in order to achieve perfect reconstruction. Is the solution $\{H_1(z), G_1(z)\}$ unique? If not, provide at least two more alternative solutions. [7]